# Research Statement

### *Advancing Generative AI via Science of Data*

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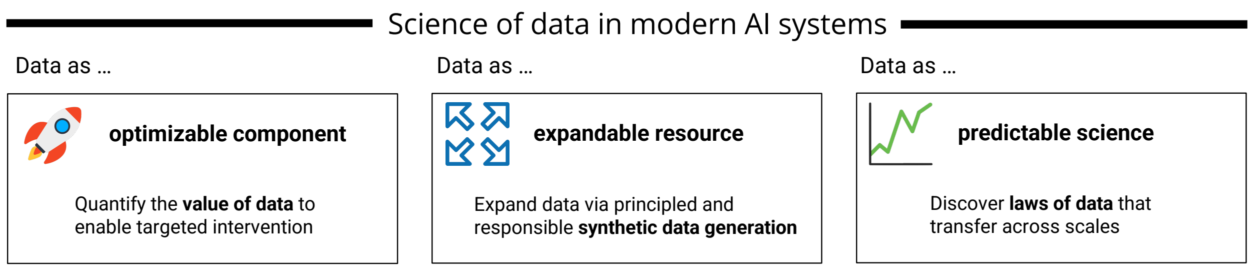
Data has fueled many of the most transformative advances in artificial intelligence (AI). The rise of generative AI further underscores its role as the driving force behind new capabilities. Yet as models and datasets reach unprecedented scales, the long-standing paradigm of assembling ever-larger corpora through trial-and-error curation is becoming increasingly unsustainable. Future progress in foundation models will depend not on scaling compute alone, but on building a principled understanding of data: how it drives learning, shapes model behavior, and defines the limits of intelligence.

I develop the **science of data**—a perspective that views data as an *optimizable*, *expandable*, and *predictable* component of modern AI systems, rather than as i.i.d. samples in statistical learning theory or static input in deep learning practice. This viewpoint offers a new path toward building AI systems that are more steerable, efficient, and capable.




## Past research

My past work investigates how data can be understood, utilized, and created within modern AI. Concretely, I establish the foundations of data attribution, develop toolkits for optimizing data use in dynamic learning systems, and design algorithms for generating privacy-preserving synthetic data that enable responsible data expansion.

**Foundations of data attribution.** Data attribution [Deng et al., 2025] quantifies the influence of individual training samples on model behavior. My work strengthens its *theoretical and algorithmic foundations* to enable more principled use of data. In Hu et al. [2024a], we study the *most influential subset selection* problem, central to assessing the robustness of inferential results across scientific domains. We show that common influence-based greedy heuristics can fail due to the non-additive nature of influence. To address this, we propose an adaptive greedy algorithm with both theoretical guarantees and strong empirical performance. In Wang et al. [2025], we show that data attribution is highly sensitive to hyperparameter choices. We then theoretically analyze the regularization term in influence function–based methods and propose a lightweight tuning strategy that achieves strong downstream performance. Together, these studies lay the foundation of data attribution in modern machine learning.

**Value of data in online reinforcement learning.** Online reinforcement learning (RL) underpins applications from games and robotics to post-training of large language models (LLMs), yet the role of data in this paradigm remains poorly understood due to the circular dependency between data and model. In Hu et al. [2025b], we take a first step toward characterizing the *value of data* in online RL. We develop a local data attribution framework based on gradient alignment, which interprets model checkpoints with respect to the recent rollout buffer and quantifies each record's contribution to the local update direction. This framework identifies records with *inaccurate advantage estimates* as a key training bottleneck and naturally motivates an algorithm that adaptively removes such harmful data, thereby accelerating learning and improving final return.

**Privacy-preserving synthetic data.** Beyond optimizing how data is used, I also investigate how to *expand* it, an emerging challenge as high-quality web corpora approach saturation. A promising direction is *privacy-preserving synthetic data* [Hu et al., 2024b], which unlocks the value of user data that would otherwise remain inaccessible due to privacy constraints and has attracted significant attention from the industry [Microsoft, 2024, Apple, 2025]. In Hu et al. [2025a] (work done while at Google Research), we propose a metadata-conditioned framework for differentially private (DP) synthetic text generation. By combining rich tabular features as metadata, a DP tabular synthesizer as feature generator, and a DP fine-tuned LLM as conditional generator, we achieve state-of-the-art results in DP text synthesis. We further introduce an RL training recipe that enhances the conditional generator's instruction-following ability. Together, these advances enhance both the quality and control of generated text under strong privacy guarantees.

**Early experiences: Data shapes machine learning theory and algorithms.** Early in my Ph.D., I focused on theoretical machine learning, where I came to recognize the central role of data. In Chidambaram et al. [2022], we show that the effectiveness of *mixup* hinges on the properties of the training data. In Hu et al. [2023a], we demonstrate that the robustness–fairness trade-off is driven by the tail behavior of the data distribution. In Hu et al. [2023b], we establish necessary and sufficient data conditions under which linear scalarization fully explores the Pareto frontier in multi-task learning. Collectively, these studies show that *assumptions* about data fundamentally determine learning behavior and theoretical guarantees, motivating my broader pursuit to understand data as a scientific object in its own right.

**Broader impact.** Beyond advancing scientific understanding, my research delivers tangible performance gains in real-world AI systems. My framework for online RL, recognized as an **Oral** at NeurIPS 2025 (top 0.3%), provides principled diagnostic tools for identifying and mitigating learning bottlenecks in online RL. It achieves 50% improvements in both sample efficiency and runtime, and was commended by the Area Chair as "*a valuable tool to the RL researcher's toolbox*." In parallel, my work on privacy-preserving data synthesis at Google Research addresses the central challenge of unlocking the value of sensitive user data. The resulting algorithm enables the creation of high-utility, differentially private synthetic data and delivers a 20% improvement in fidelity over prior state of the art. It is now being integrated into production pipelines to support Google's billion-user products, including Gmail's smart writing features.

## Future directions

As AI systems such as large language models continue to expand in scale and capability, understanding and improving the use of data has become both more challenging and more essential. Building on my past work, which has focused on optimizing and expanding data in learning systems, I aim to further advance this perspective by uncovering general principles that make data behavior *predictable*. To this end, I plan to pursue three interconnected directions: investigating the value of data in LLM post-training, developing feedback-driven pipelines for synthetic data generation, and uncovering the laws of data at scale.

**Data as optimizable component — value of data in LLM post-training.** Post-training is the stage where LLMs acquire higher-order abilities (e.g., alignment, instruction following, and multi-step reasoning) through processes including supervised fine-tuning (SFT) and reinforcement learning with human or verifiable feedback. As training algorithms become standardized, further progress will depend more on effective use of data. While pre-training has benefited from well-studied data curation pipelines, our understanding of how data shape model capabilities in post-training remains limited. Unlike pre-training, which operates under a fixed paradigm of next-token prediction with a cross-entropy loss, post-training presents distinct challenges. It optimizes objectives beyond language modeling, proceeds through multiple learning stages (SFT followed by RL), and exhibits strong curriculum effects that tightly couple data *ordering* with learning dynamics.

My goal is to build a framework that quantifies and optimizes the value of data throughout post-training. Building on my prior work on local data attribution for online RL, I am developing methods that leverage *gradient signals* to guide prompt selection that adapts to the model's current training dynamics, offering a principled alternative to difficulty-based heuristics. In parallel, I am exploring how to design attribution targets in SFT that better capture genuine capability gains rather than superficial improvements in token likelihood, and how these choices interact with RL to shape learning outcomes. Together, these efforts aim to make post-training data an actively optimized component of the learning process, enabling more targeted and controllable model improvement.

**Data as expandable resource — feedback-driven synthetic data generation.** Optimizing how existing data is used is an important step, but as web-scale corpora approach saturation, sustaining long-term progress demands creating data that can extend model capabilities beyond what existing datasets provide. Synthetic data offers a promising direction, yet current approaches remain largely heuristic and inefficient.

I aim to integrate *data attribution* into a closed-loop framework where attribution signals directly guide data creation. These signals can detect low-quality synthetic samples, enabling selective filtering that improves data quality and training stability. They can also reveal capability gaps, guiding targeted generation that broadens coverage and mitigates collapse. As models evolve, attribution enables us to track how the value of synthetic data changes, providing feedback to dynamically balance real and synthetic data. Critically, these signals must be *adaptive* and anchored to the real world—through user feedback, back-tested validation on future text, and other external signals—to avoid overfitting to a fixed validation set. Together, these efforts aim to evolve synthetic data pipelines from heuristic engineering into feedback-driven systems that continually refine their own training signals.

**Data as predictable science — laws of data at scale.** Scaling pushes modern AI systems into regimes where brute-force data practices are not sustainable. To design data effectively at such scales, we must understand when insights derived at smaller settings can be extrapolated to larger ones. Yet current tools such as data attribution provide only a *static* view with respect to a fixed model and dataset, but fail to capture how these relationships evolve with model capacity and training distributions.

I aim to investigate the scaling behavior of data, in analogy to neural scaling laws describing how performance trends with compute and model size. Specifically, I will study how quantities such as *individual data value*, *inter-sample interactions*, and *optimal data composition* vary across scales. Patterns that remain consistent enable data decisions to be derived from lightweight experiments, avoiding costly large-scale training. For patterns that vary with scale, I aim to develop predictive models that characterize how data behavior changes across regimes. Ultimately, my goal is to uncover transferable and predictable regularities in how data shapes learning, offering principles to guide the design and optimization of large-scale AI systems.

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